

Short Circuit Bolted-Fault Analysis of LVDC Power Distribution Systems for Datacenter Uses

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Presentation flow

Introduction – why this study was conducted	01
Modeling assumptions and system overview	02
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Modeling results – observed energy levels and waveform characteristics	04
Conclusion – impact and where to go from here	05

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Introduction

- Low-voltage direct current (LVDC) distribution systems are becoming increasingly common in modern power systems due to their:
  - Improved efficiency
  - Simplified conversion stages
  - Reduced conductor requirements
- This work was developed in support of ongoing industry efforts to better understand protection and fault behaviors in emerging OCP-aligned 800 VDC architectures.
- This study examines the electrical factors that influence fault severity and provides insights to:
  - Help improve protection strategies
  - Guide equipment design
  - Support future testing and standards development



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Modeling assumptions

1

Bus voltage

800 VDC conductor-to-conductor; single reference node—not a ±400 V midpoint system

2

Stored energy

Equivalent bus capacitance: 2 mF, 5 mF & 10 mF; 1 mF, 78 mF, 3136 mF, & 23,521 mF used for supercapacitor sizing

3

Distribution path

Copper conductor size: 8.4 mm², 42.5 mm², 150 mm² and 240 mm²; lengths: 0.25 m, 2 m, 5m, & 10 m

4

Geometry and temperature

10 mm center spacing for smaller conductors; 25 mm for larger; resistance at 20°C, 40°C, & 70°C

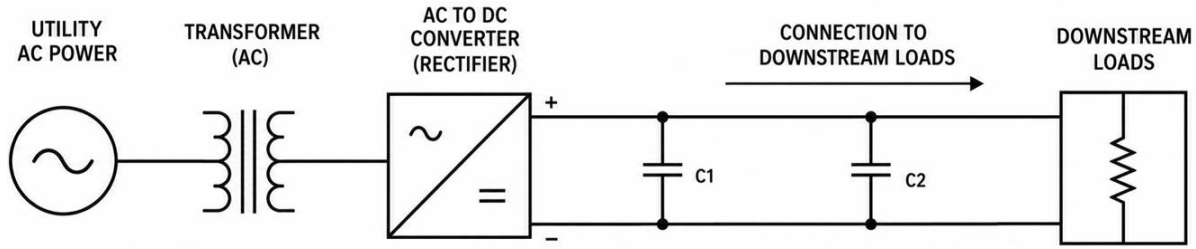


Figure 1

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Simplified system models

The previously calculated and estimated parameters were used to simulate both models.

- **Model 1** (Figure 3) strictly focuses on system capacitance, loop inductance and loop resistance.
- **Model 2** (Figure 4) incorporates a DC/DC converter to emulate power conversion with supercapacitors at the output.

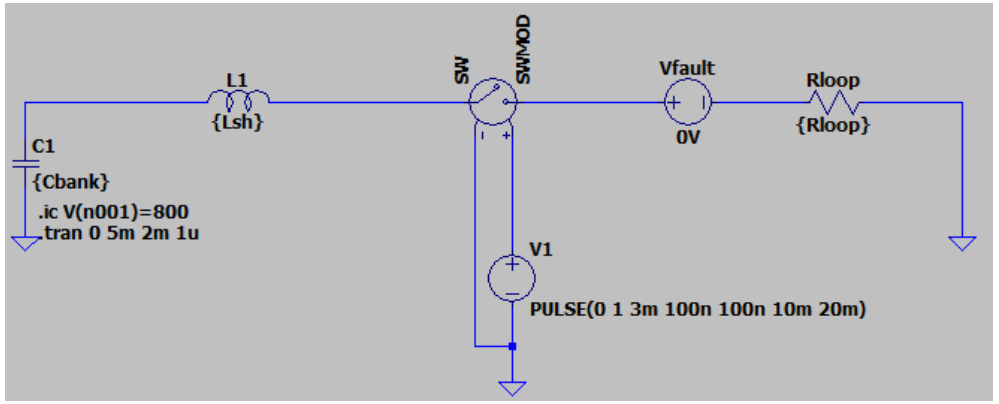


Figure 3

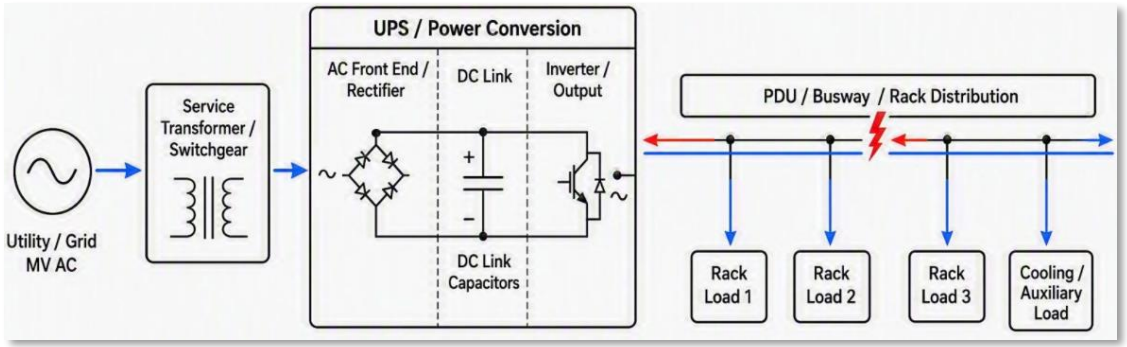


Figure 2

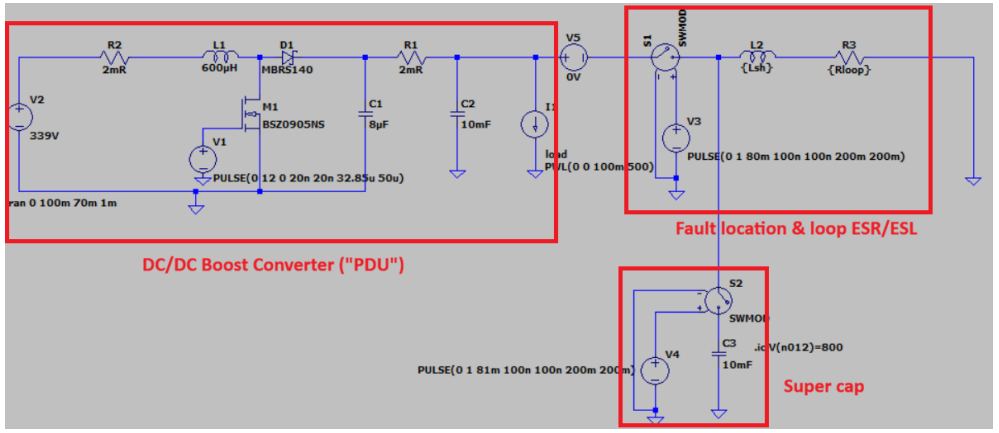


Figure 4

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Simulated Waveforms for Model 1 – 5 m & 5 mF @ 20°C

- Oscillatory behavior was observed due to the loop resistance being lower than the inductive and capacitive elements which created an underdamp scenario.
- As conductor size increased, loop resistance and inductance decreased, which resulted in greater peak currents.

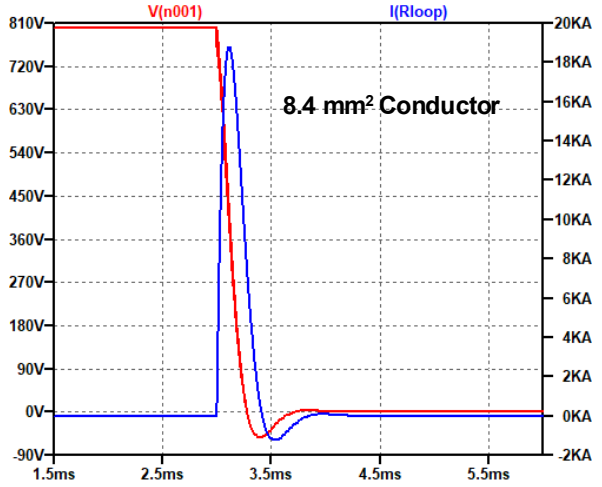


Figure 5

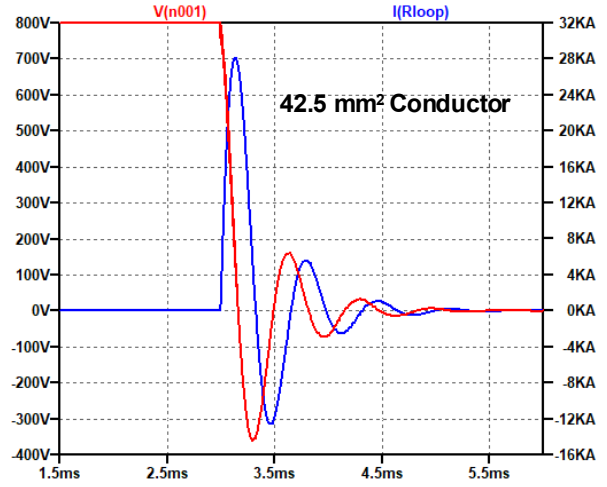


Figure 6

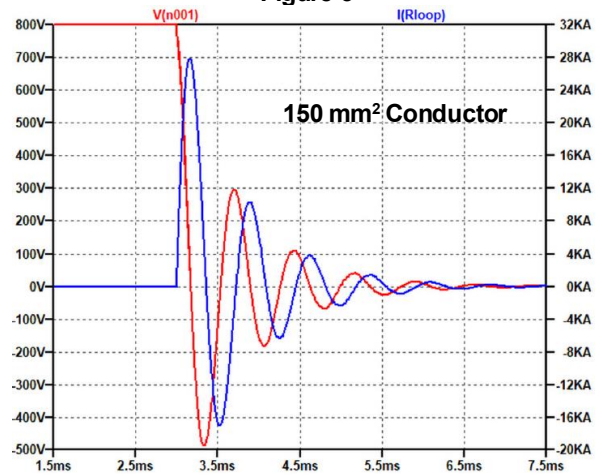


Figure 7

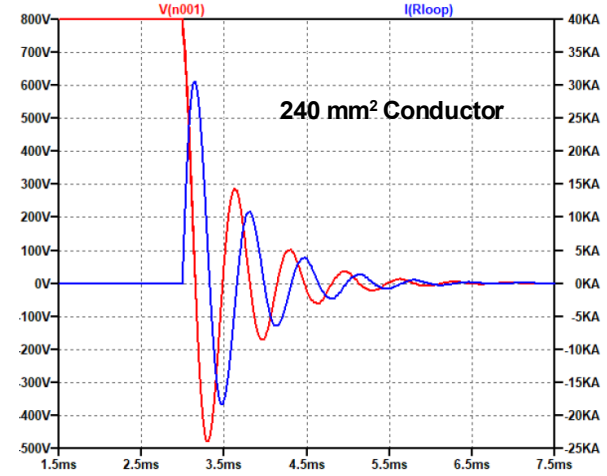


Figure 8

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Full simulated results for Model 1

8.4 mm² Conductor						
Length (m)	20°C Ambient			70°C Ambient		
	Pk I (kA)	I²t (A²s)	di/dt (kA/µs)	Pk I (kA)	I²t (A²s)	di/dt (kA/µs)
0.25	63.3	227,563	2.05	62.5	221,269	2.05
2	27.6	112,589	0.33	26.3	101,131	0.32
5	16.3	60,329	0.13	15.2	52,354	0.13
10	10.3	34,005	0.06	9.4	29,032	0.06

Table 2

42.5 mm² Conductor						
Length (m)	20°C Ambient			70°C Ambient		
	Pk I (kA)	I²t (A²s)	di/dt (kA/µs)	Pk I (kA)	I²t (A²s)	di/dt (kA/µs)
0.25	75.1	258,032	3.09	74.8	256,369	3.09
2	41.6	209,947	0.61	40.9	201,487	0.61
5	28.2	159,031	0.26	27.5	147,452	0.26
10	20.1	113,386	0.13	19.4	101,839	0.13

Table 3

150 mm² Conductor						
Length (m)	20°C Ambient			70°C Ambient		
	Pk I (kA)	I²t (A²s)	di/dt (kA/µs)	Pk I (kA)	I²t (A²s)	di/dt (kA/µs)
0.25	72.5	263,983	2.64	72.4	263,548	2.64
2	40.2	247,640	0.50	40.0	244,239	0.50
5	27.9	223,743	0.21	27.8	216,772	0.21
10	20.6	192,744	0.11	20.4	182,834	0.11

Table 4

240 mm² Conductor						
Length (m)	20°C Ambient			70°C Ambient		
	Pk I (kA)	I²t (A²s)	di/dt (kA/µs)	Pk I (kA)	I²t (A²s)	di/dt (kA/µs)
0.25	75.5	264,857	3.02	75.5	264,857	3.02
2	43.4	254,333	0.60	43.3	252,329	0.60
5	30.6	239,061	0.26	27.8	233,203	0.26
10	22.8	215,020	0.13	22.6	207,224	0.13

Table 5

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Simulated waveforms for Model 1—varying capacitance

- Capacitance variations of 2 mF, 5 mF, & 10 mF were modeled while keeping conductor length and size constant.
- Higher capacitances yielded greater peak currents at a longer time with less oscillations.
- Lower capacitance values created faster transients with lower peak currents and more oscillations.

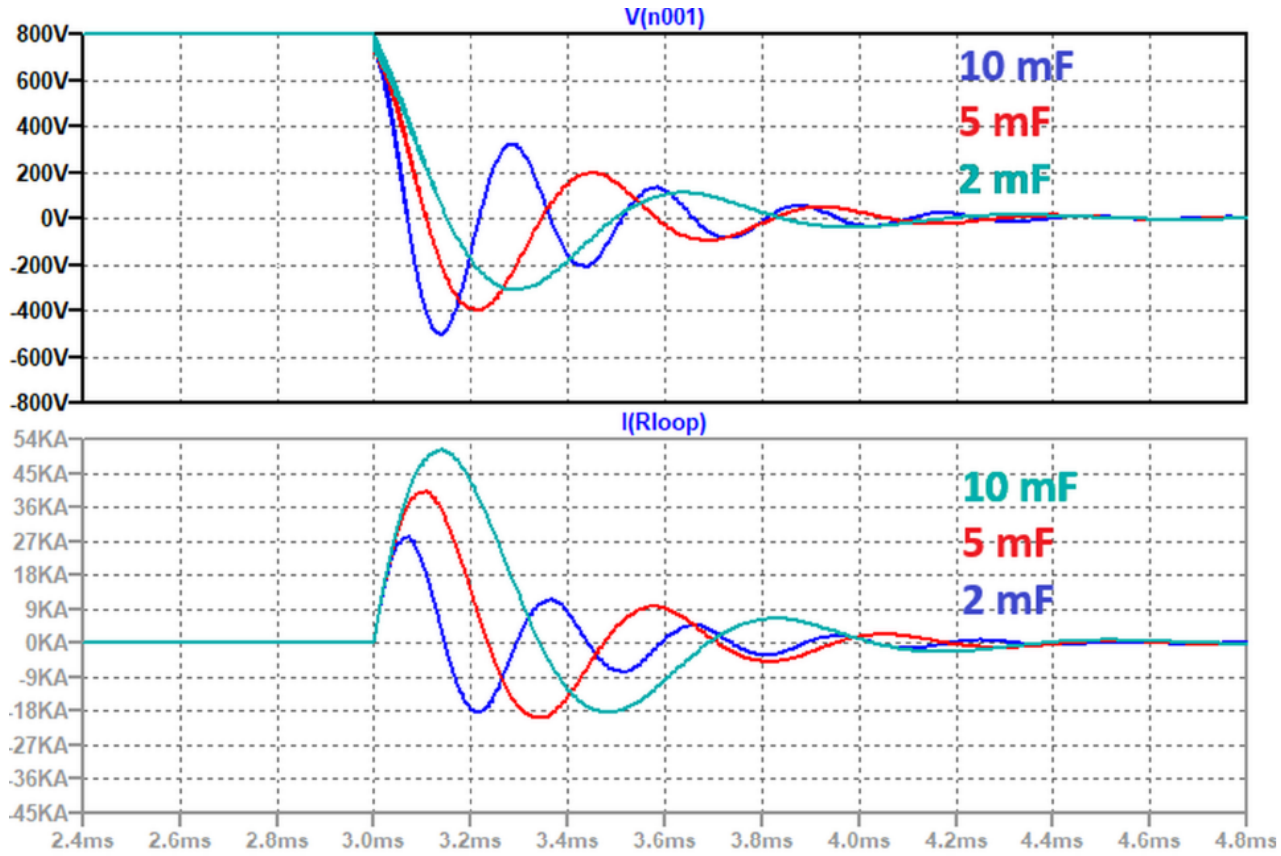


Figure 9

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Calculated supercapacitor values for Model 2

- Supercapacitor values were calculated using the following:
  - $U_C = U_0 \cdot e^{-t/RC}$
- U_C is the minimum allowable bus voltage
- U_0 is the starting voltage
- t is the required support duration
- R & C are the resistance & capacitance
- Solving for C:
  - $C = \frac{-t}{R \cdot \ln(\frac{Uc}{U0})}$
- Final C values: **1 mF, 78 mF, 3,136 mF, & 23,521 mF**



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Simulated waveforms for Model 2

- Supercapacitors significantly increased the magnitude of fault currents.
- Peak currents ranged from 8 - 70kA
- I^2t ranged from 1.3 - 689.5 kA^2s
- di/dt values ranged from 0.03 - 0.15 kA/µs

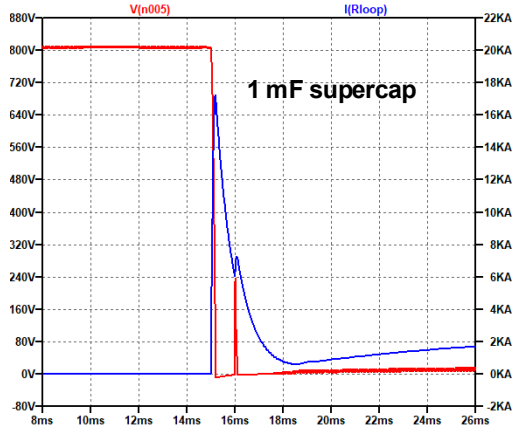


Figure 10

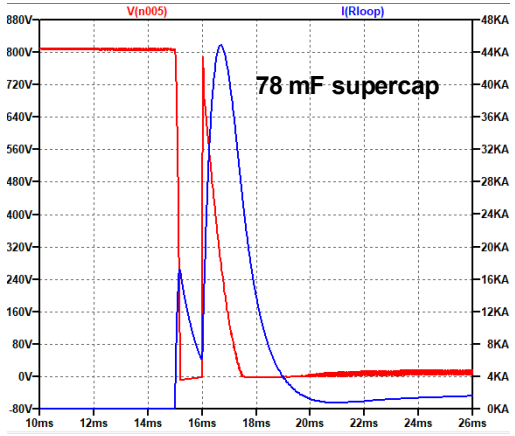


Figure 11

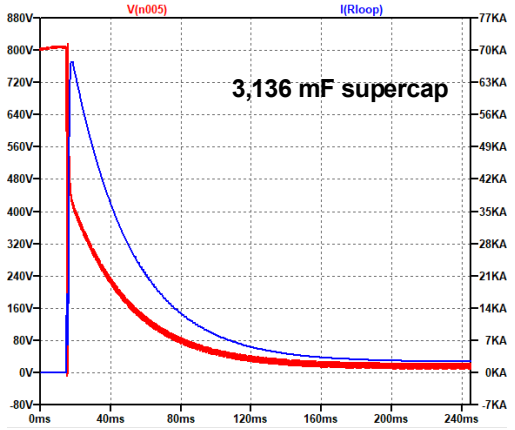


Figure 12

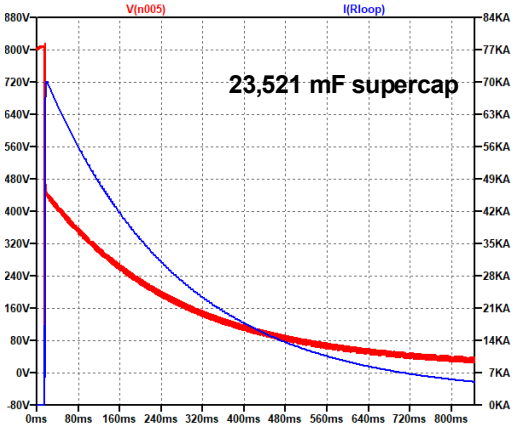


Figure 13

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Conclusions

- The baseline study showed:
  - Stored energy and loop impedance set fault severity.
  - Large energy-storage branches can dominate total fault energy, current rise rate and duration.
  - Low damping can create reverse current, oscillations and secondary peaks.
  - Rapid current rise and elevated peak values can exceed the interrupting capabilities of protection devices.
- Future work: incorporating realistic converter & load behavior, improve system impedance representation, evaluate protection technologies, and provide technical data for standard development.

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Call to action!

- Participate in codes and standards development – use validated data to inform OCP workstreams and future UL/IEC requirements.
- Share real system data to help improve fidelity of models and tests.
- Help identify representative fault scenarios for bolted faults, fast interruption, converter faults, energy-storage contribution, and transient over-voltages.
- Support hardware validation by comparing simulated results with measurements from representative equipment.
- Find additional information including white papers:
  - ul.com/industries/data-centers
- Get in touch & consider joining our task group of engineers!
  - Sarkis.juana@ul.com, Danish.Zia@ul.com, Dirk.Mueller@ul.com

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